

EARLY WARNING SYSTEM

FINTECH FINAL PROJECT : GROUP 2

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THE PROBLEM

One of the biggest problems of financial markets is that they tend to crash

Market crisis correspond to "risk-off" situations, in which risk premia and financial assets exhibit anomalous behavior

Rather than predicting risk-off situations, it is sufficient to recognize them at their dawn

Main goal: detecting crisis before most damage has been made and reducing false alarms

THE PROCEDURE

Analysis of the dataset and inspection of its features



Face the problem using different approaches



Analysis of the results and exploitation of them to build a Toy Portfolio

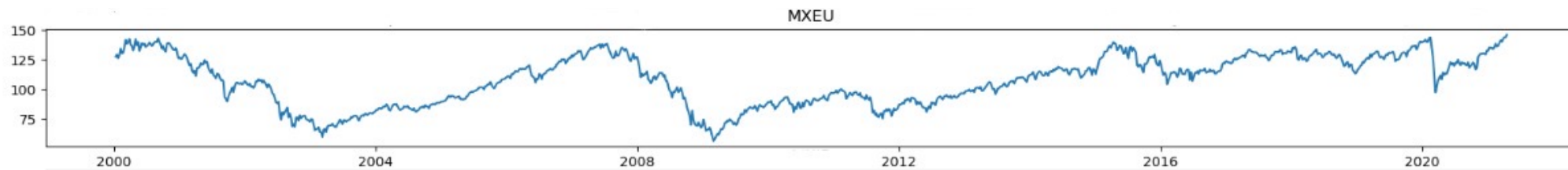
Logistic Regression
Random Forest
Copula
Autoencoder

DATASET INSPECTION

Analysis of the plot of the times series of each feature in order to check the stationarity.



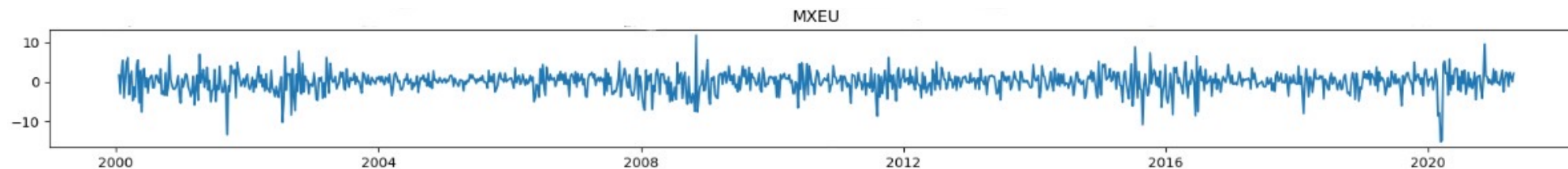
Non stationarity



First Differences technique applied to ensure the stationarity of the data.



Graphical check and Dickey-Fuller test



MODEL 1: LOGISTIC REGRESSION

Split the dataset using an outlier detection approach.

Training set



Ridge Regression training to avoid overfitting problems.

Validation set



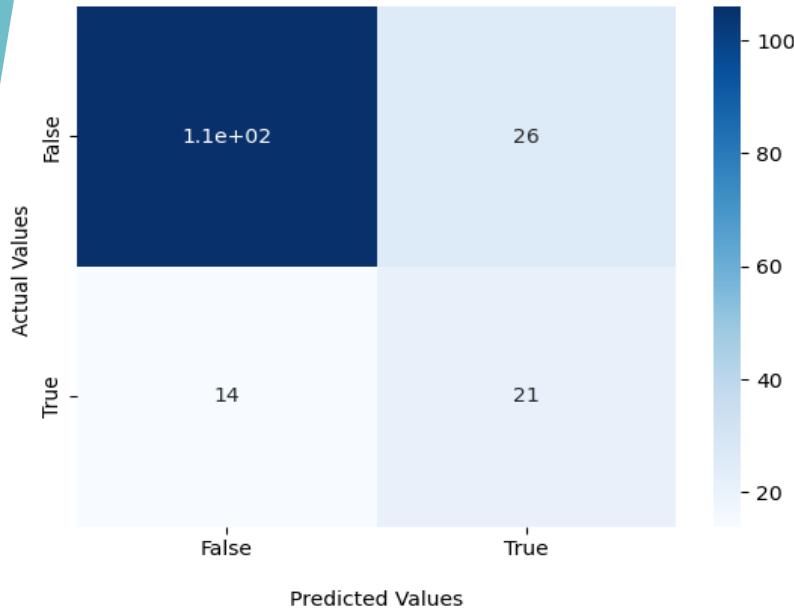
Optimization of the best threshold with respect to the best F1 score.
The threshold is the minimum value of the probability predicted to be assigned to class '1'.

Test set

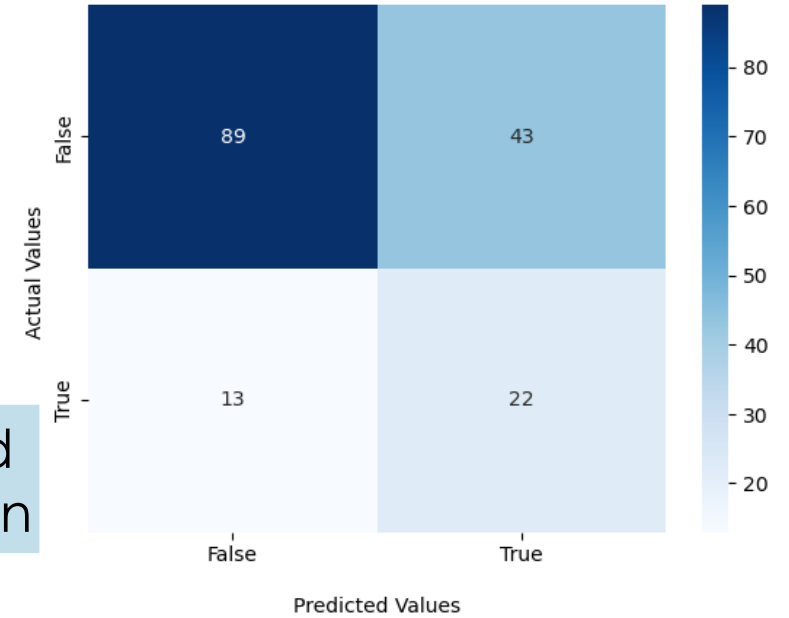


Evaluation of the results.

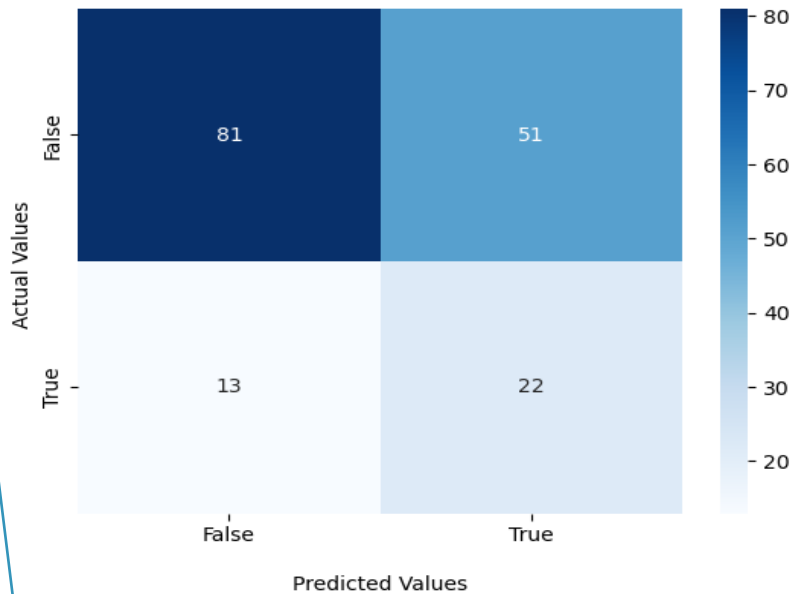
CONFUSION MATRICES



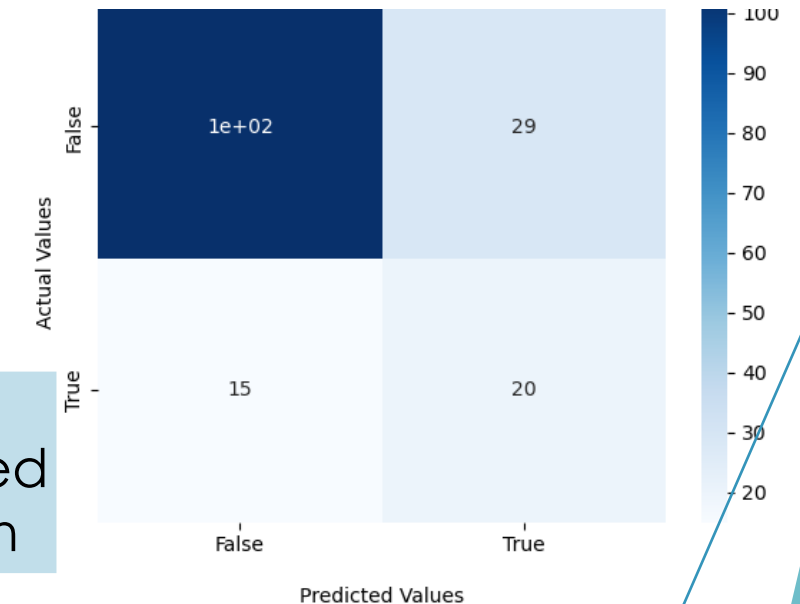
Ridge
Regression



Weighted
Regression



Smote
Regression

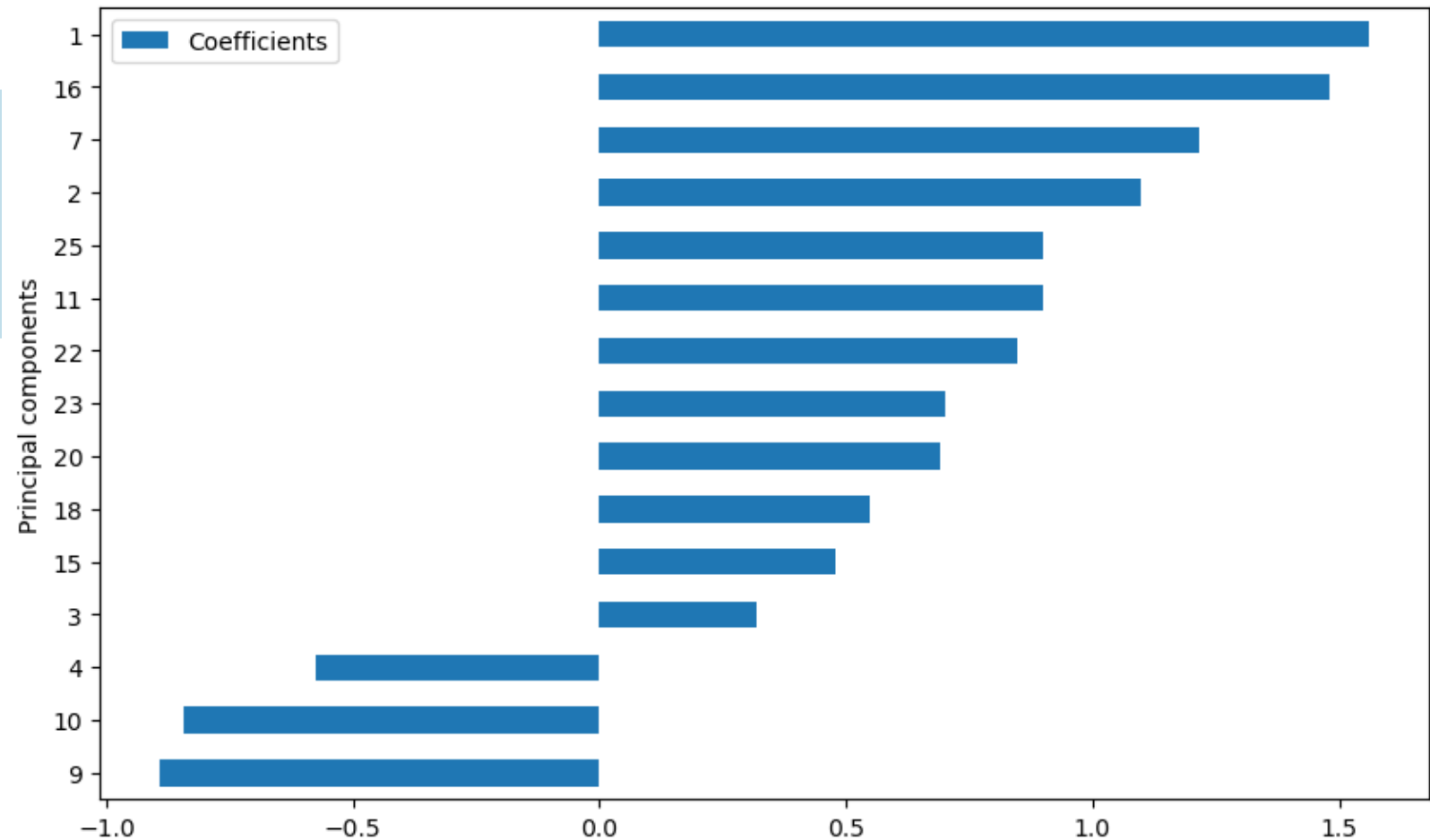


Optimal
Regularized
Regression

REGRESSION WITH PCA

PCA allows to have orthogonal features and with 25 regressors it is possible to explain the 95 % of the variance.

The graph takes into account only the significant coefficients



PCA INTERPRETATION

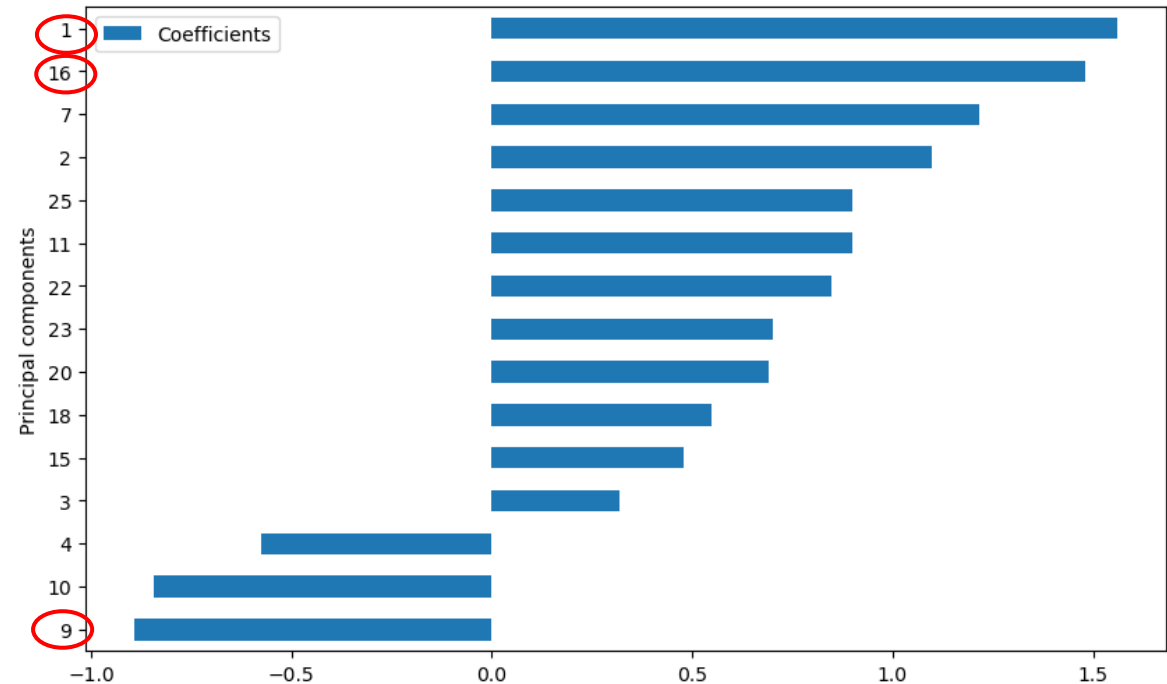
PRINCIPAL COMPONENT	POSITIVE EFFECT	NEGATIVE EFFECT
1	VIX	GT10, USGG30YR, GTDEM2Y, GTDEM10Y
16	XAU BGNL	GTJPY30YR
9	GTDEM2Y, USGG2YR, GTGBP2Y	-

Accuracy = 0.75

Precision = 0.43

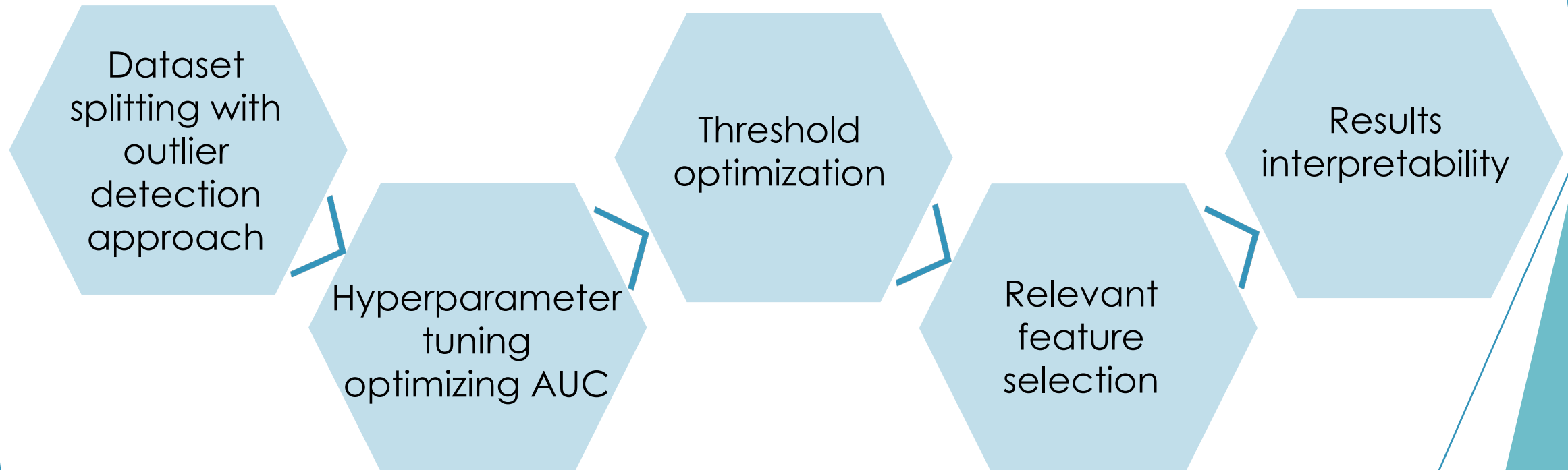
Recall = 0.57

F1 score = 0.49



MODEL 2 : RANDOM FOREST

- From the previous results it is evident that the logistic regression is too simple, even if it allows us to gain deep insights on our features.
- We try to improve the predictive capability with a black box model.



FEATURE SELECTION

AUTOMATIC

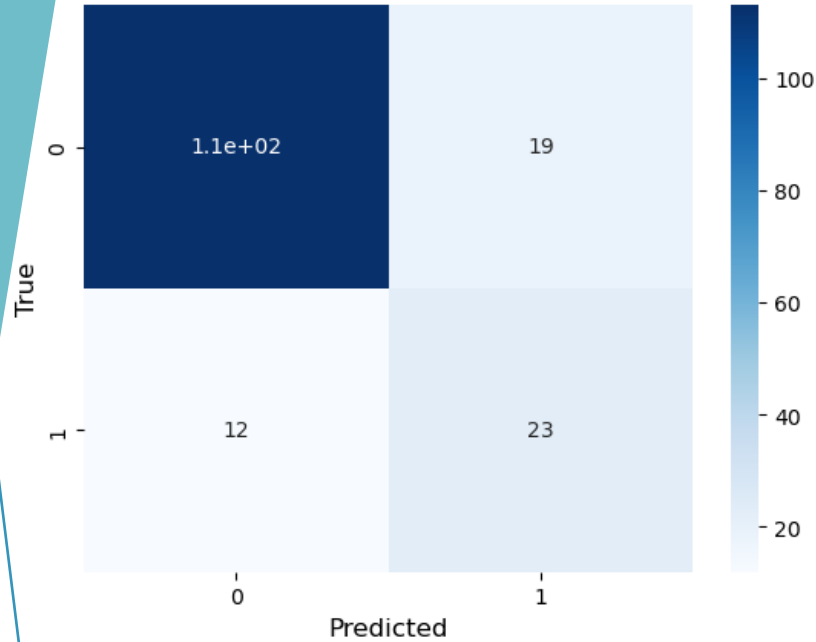
- MXEU
- MXUS
- LP01TREU
- VIX
- GTDEM2Y
- BDIY
- GTITL2YR
- LF98TRUU
- USGG30YR
- US0001M

BUSINESS ORIENTED

- MXEU
- MXUS
- LP01TREU
- VIX
- GT10
- MXJP
- MXBR
- MXCN
- MXRU
- MXIN
- GTDEM10Y

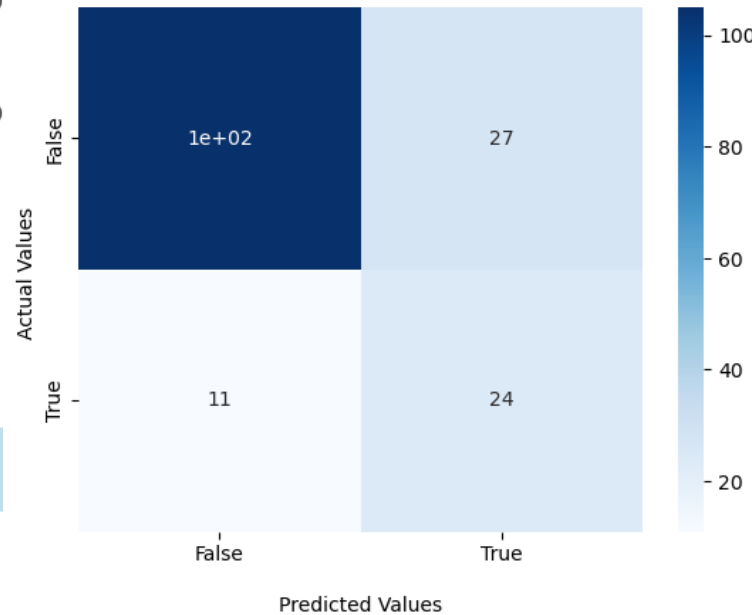
CONFUSION MATRICES

Full dataset



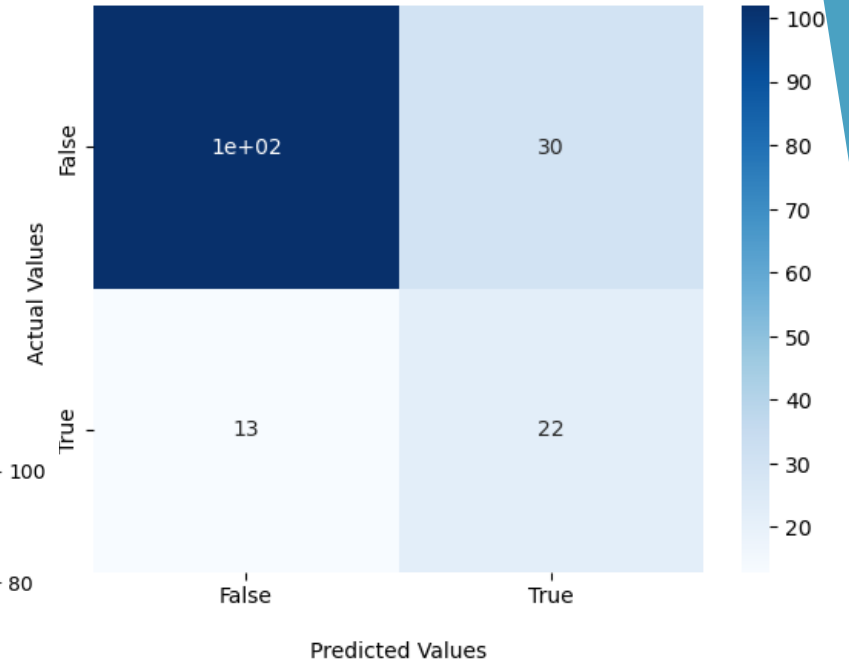
AUC = 0.832

Automatic selection



AUC = 0.804

Business oriented selection

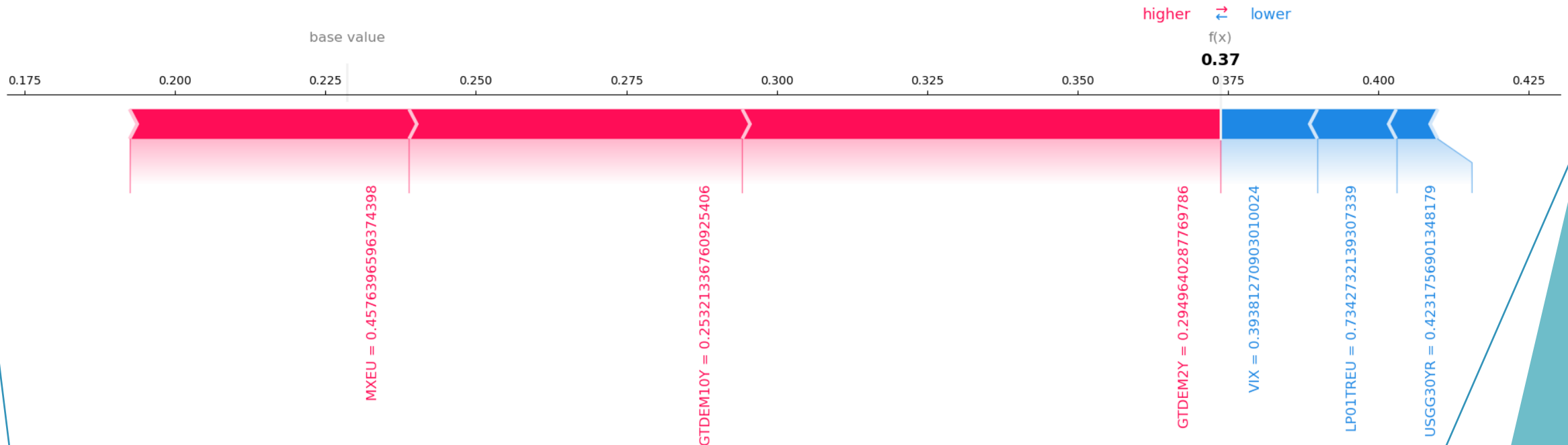


AUC = 0.757

LOCAL INTERPRETABILITY OF THE RESULTS

TRUE NEGATIVE :

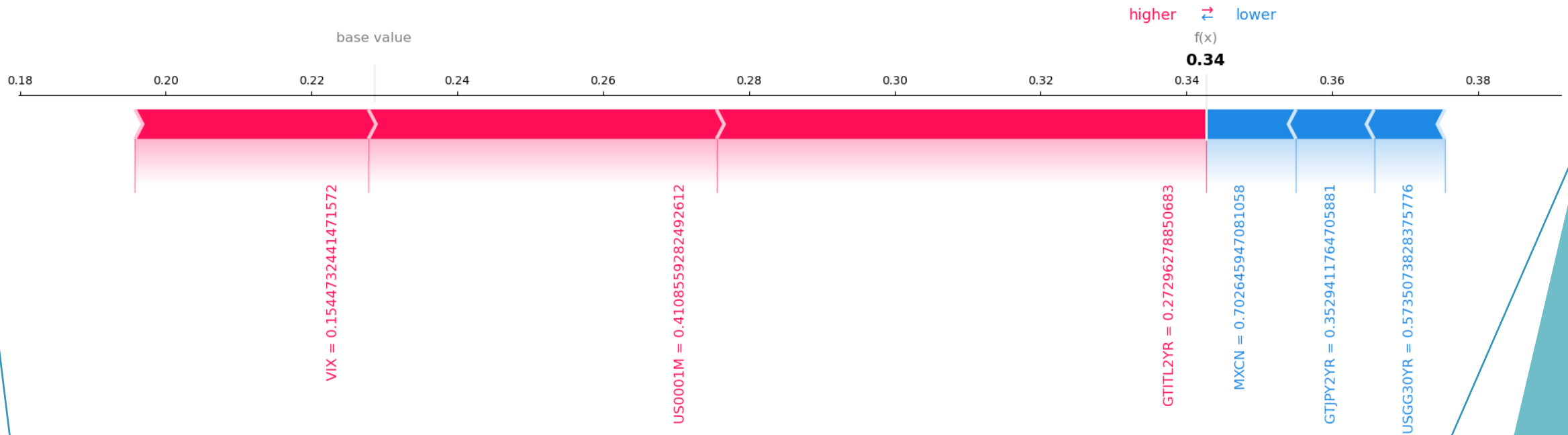
The most significant features for prediction of this true negative observation are : MXEU and GTDEM10Y and GTDEM2Y that increase the probability of being a normal period sample, while instead the others (for example VIX) contribute negatively.



LOCAL INTERPRETABILITY OF THE RESULTS

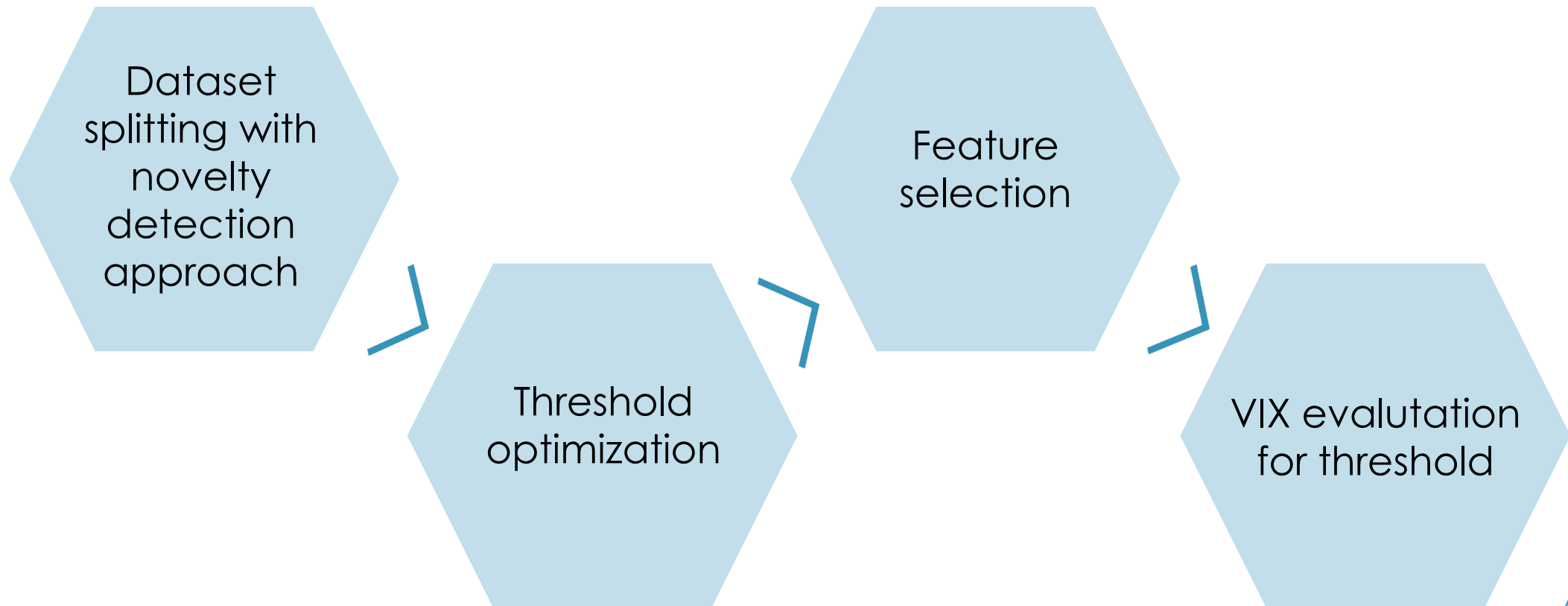
TRUE POSITIVE:

The most significant features for prediction of this true positive observation are : VIX and US0001M and GTITL2YR that increase the probability of being an abnormal period sample, while instead the others (for example MXCN) contribute negatively.



MODEL 3 : GAUSSIAN COPULA

- In this case we try to implement a probabilistic model.



MODEL 3 : GAUSSIAN COPULA

Training set



Training of a Gaussian copula.

Validation set



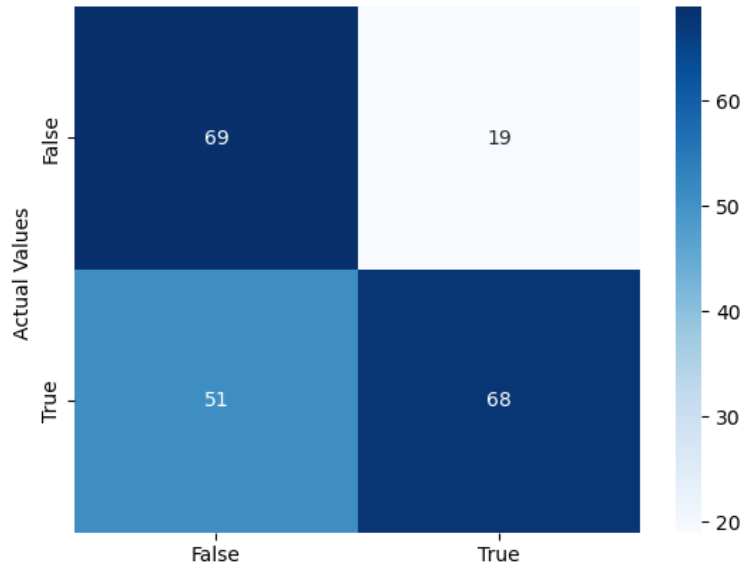
Optimization of the best threshold with respect to the largest AUC and with respect to the F1-Beta score.

Test set

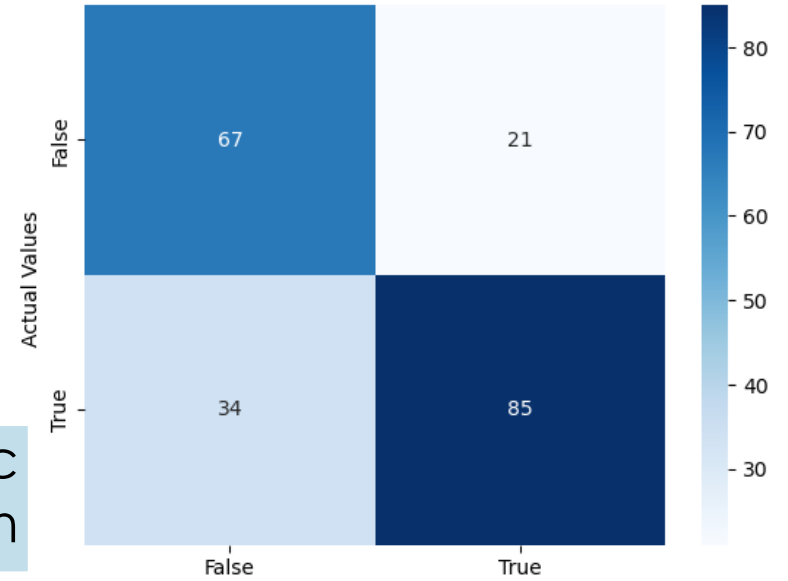


Evaluation of the results in the two cases.

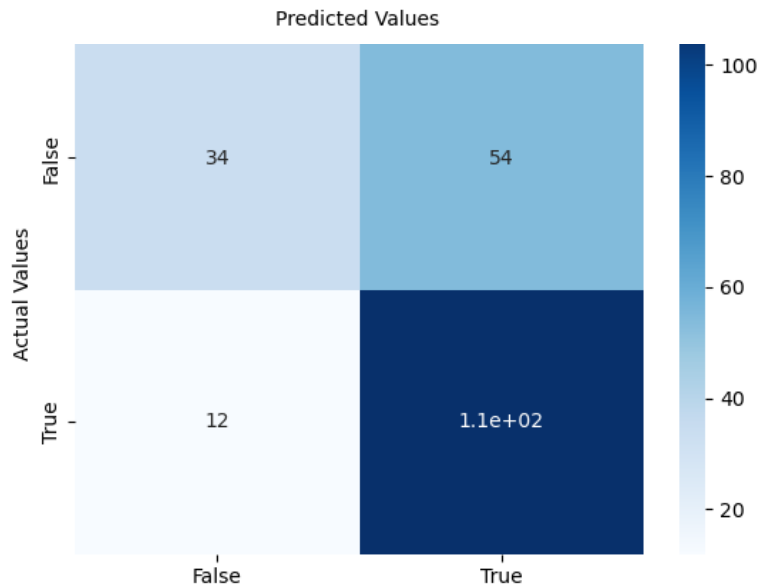
CONFUSION MATRICES



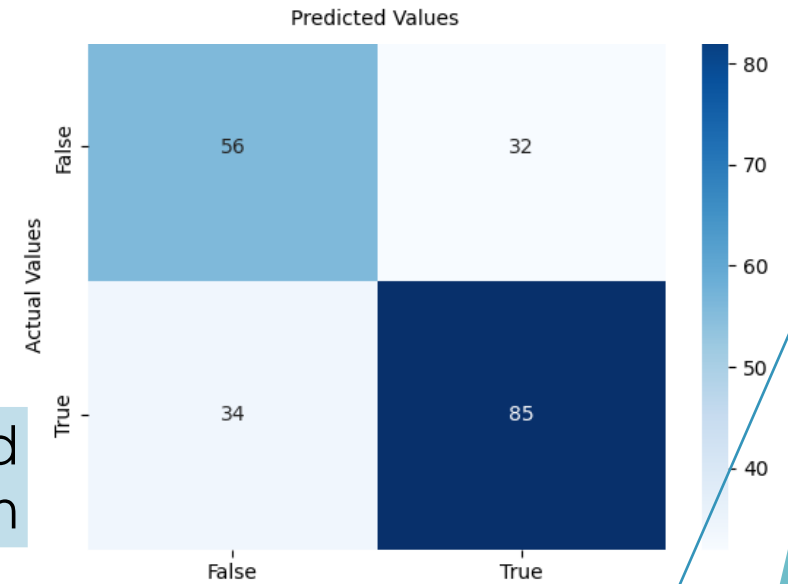
AUC
Optimization



Automatic
feature selection



F1-Beta
Optimization

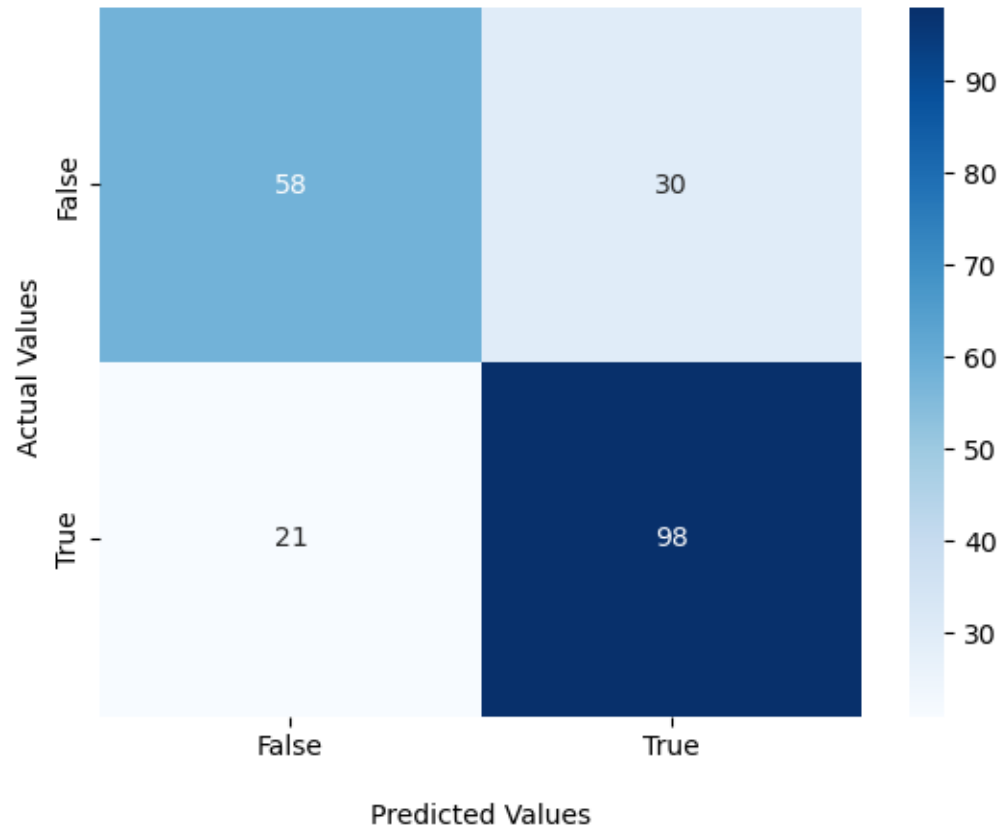


Business oriented
selection

VIX EVALUATION FOR THRESHOLD

In order to improve the model predicting capabilities, we exploit business knowledge about the feature VIX, that during crisis periods becomes very large. We can make use of this by adding a further condition on the predictions.

These are the results :



Accuracy = 0.75

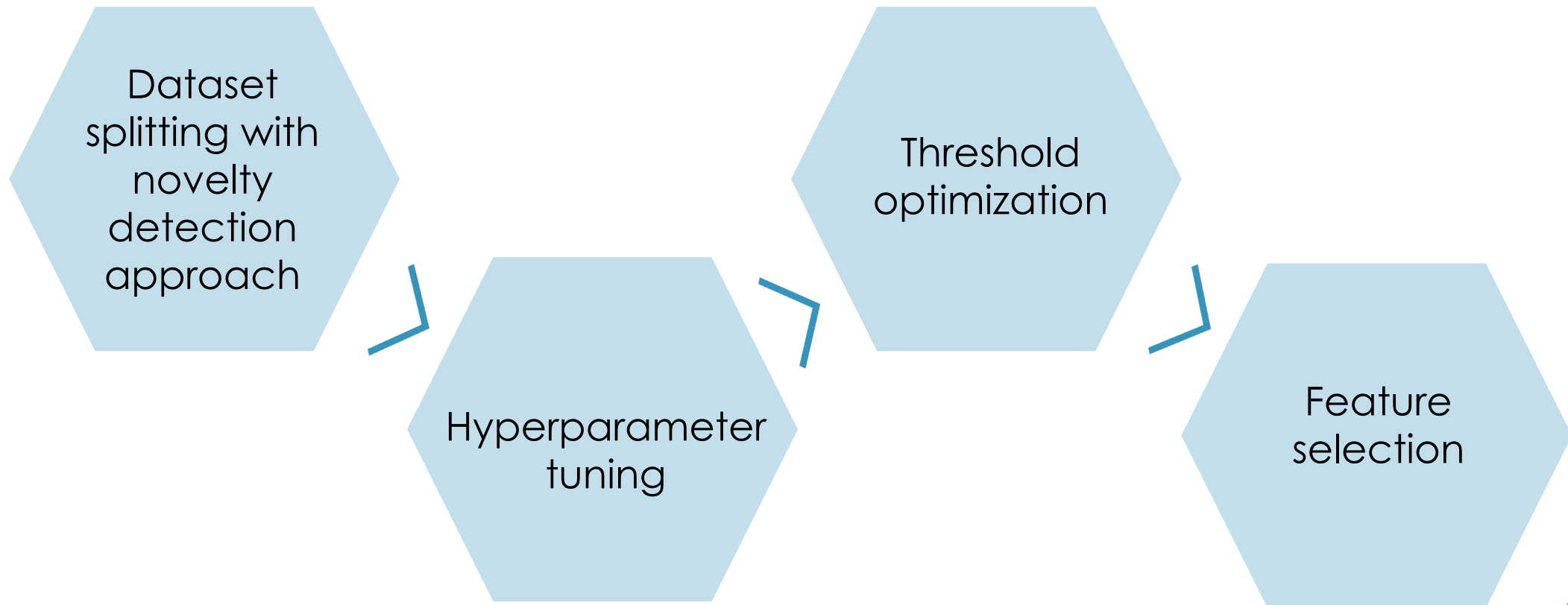
Precision = 0.77

Recall = 0.82

F1 score = 0.79

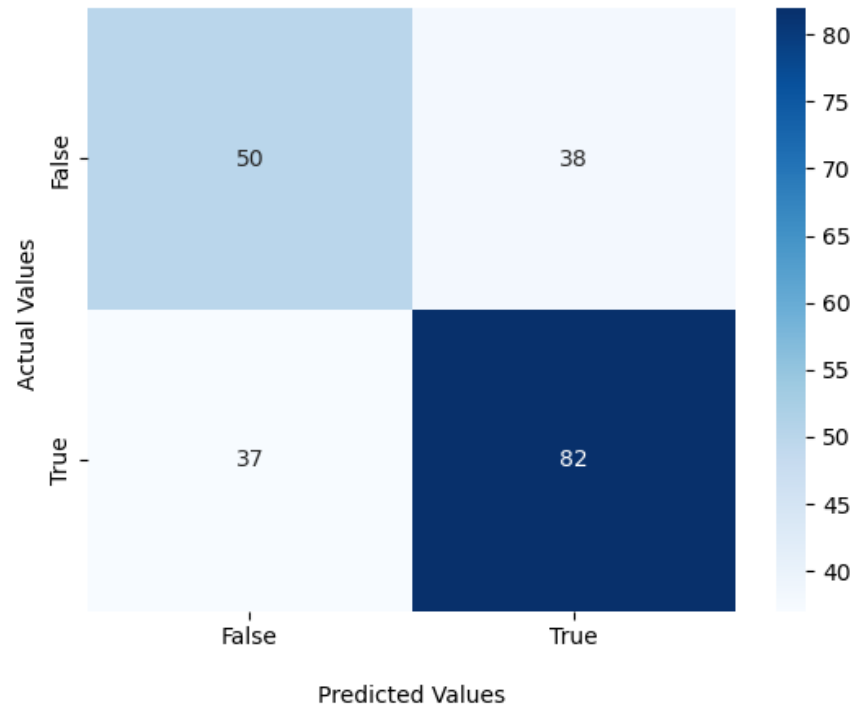
MODEL 4 : AUTOENCODER

- The Autoencoder is a neural network model that exploits dimensionality reduction.
- Its performances are evaluated on the reconstruction error.

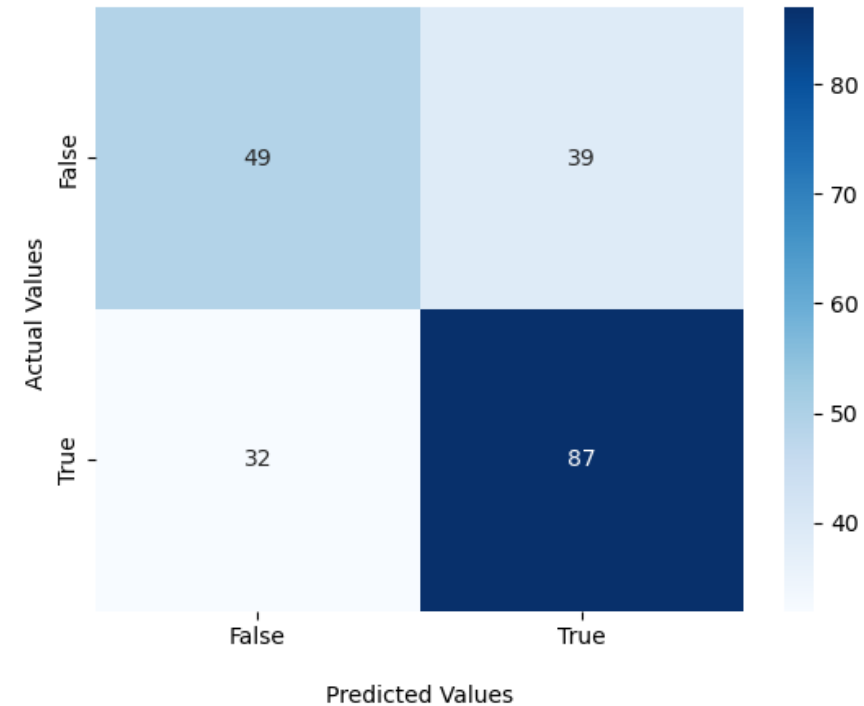


MODEL 4 : AUTOENCODER

The structure of the neural network is composed by 3 hidden layers with respectively 8, 4 and 8 neurons. We choose as hyperparameters a learning rate of 0.0001 for the Adam optimizer and a number of epochs of 800. The latter is chosen after a comparison of the MSE loss in training and validation to ensure no overfitting.



Complete model



Business oriented
selection

TOY PORTFOLIO

We try to simulate a portfolio buying some of the indexes of the dataset, rebalancing it weekly, according to the detection of risk-on and risk-off periods obtained with our models.



The weights are selected following this procedure:

- Diversified weights in all the assets during normal periods;
- Long positions only on safe assets Gold and VIX, during abnormal ones.



Comparison of the toy portfolio's performances with the benchmark S&P 500 Total Returns Index's ones, including dividends reinvesting, in the period 2011-2021.

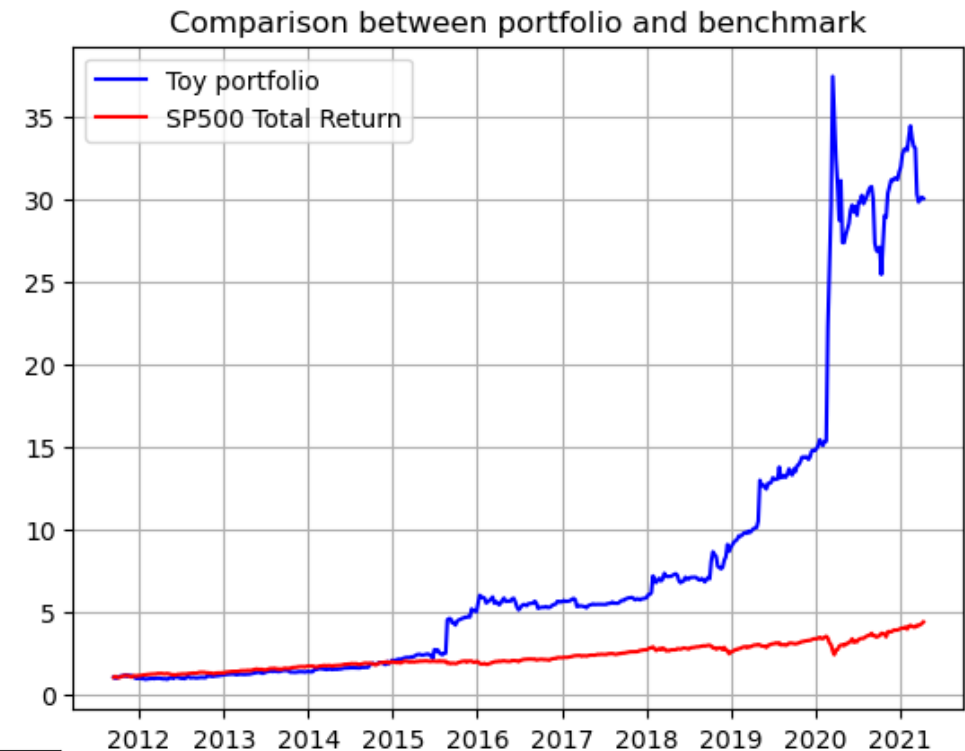
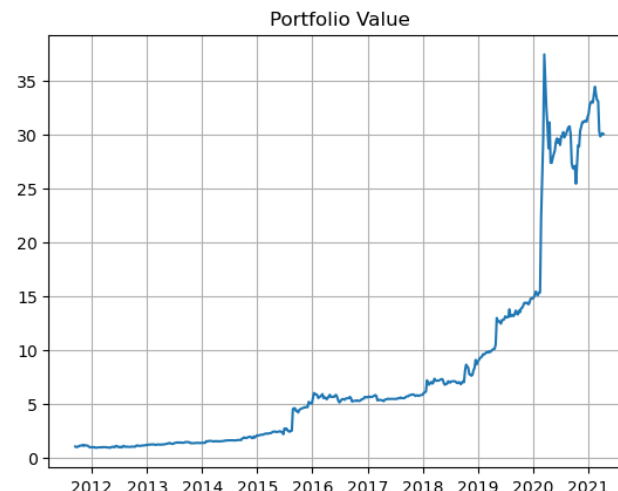
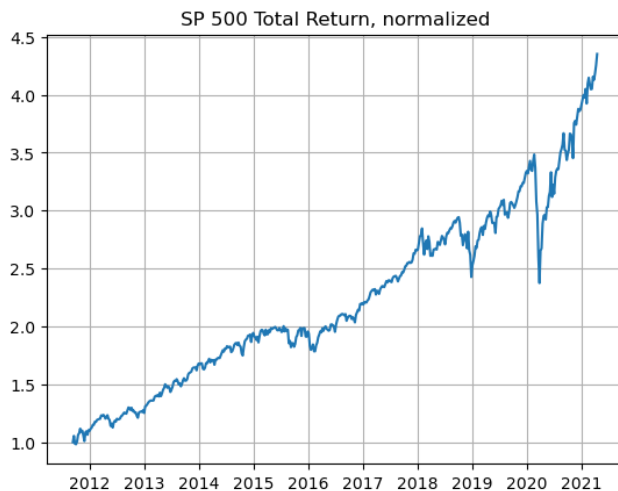
ASSETS

- Gold spot
- Dollar Index spot
- Volatility Index
- MSCI USA
- MSCI Europe
- MSCI Japan
- MSCI Russia
- MSCI China

TOY PORTFOLIO: RANDOM FOREST

We obtain very high returns since the model is able to detect crisis periods and the most evident is the abnormal period in 2020.

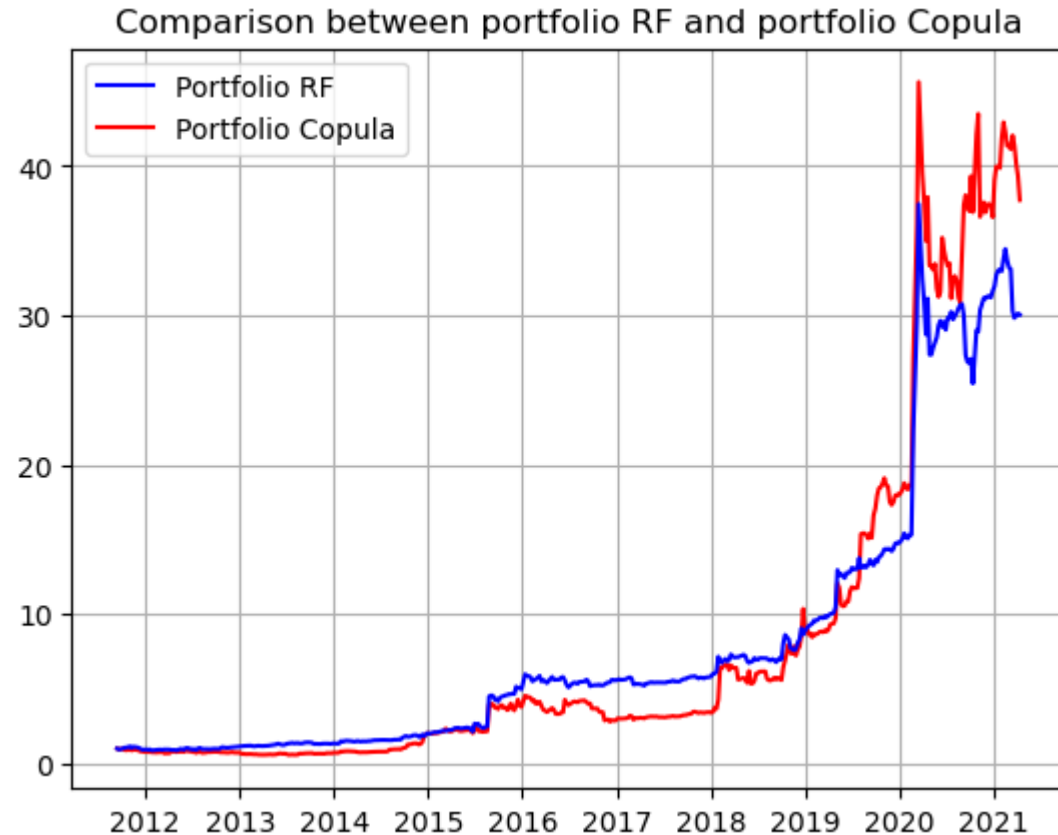
We are not considering the presence of transaction costs when rebalancing the portfolio, this would decrease the overperformance. Nevertheless we are satisfied of the result.



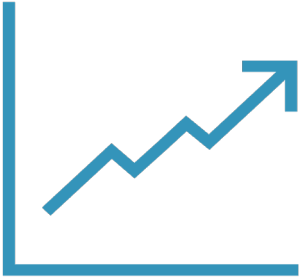
TOY PORTFOLIO: COPULA

Also the detection obtained through the Gaussian Copula model with the features selected by the Random Forest, and the additional constraint on VIX, provides good results.

We represent a comparison between the two portfolios. We can observe that they follow the same trend.



CONCLUSIONS



Trying different models we have the best performances with Copula and Random Forest. Indeed their application on the toy portfolio returns good results.

Moreover Random Forest allows us the selection of the most relevant features.



Logistic Regression is a too simple model, but the interpretation of the results is straightforward.

The autoencoder probably needs a larger quantity of data to perform better.

GROUP 2



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Thanks for your attention!